

7.H2 Law of Cosines

Lesson Introduction

 Benchmark	MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.		 Learning Target	I can solve mathematical and real-world problems that involve triangles using the Law of Cosines.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	How can you solve mathematical and real-world problems that involve triangles using the Law of Cosines?			
 Lesson Narrative	In this lesson, students will learn about the Law of Cosines and how it can be used to solve both mathematical and real-world problems. Students will begin by collaborating to derive the formula for the Law of Cosines. They will then learn how to use the Law of Cosines as a means to solve problems.			
 Component Summary	7.H2.1 Warm-Up (~5 min.)	Reasoning if two triangles are similar (<i>SE p. 434</i>)	7.H2.4 Guided Practice (10 min.)	Finding measures using the Law of Cosines (<i>SE p. 437</i>)
	7.H2.2 Collaboration (10-15 min.)	Deriving the formula for the Law of Cosines Collaboration activity (<i>SE p. 434</i>)	7.H2.5 Your Turn! (5-10 min.)	Finding measures using the Law of Cosines (<i>SE p. 438</i>)
	7.H2.3 Guided Instruction (~10 min.)	Finding measures using the Law of Cosines (<i>SE p. 436</i>)	7.H2.6 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 439</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Law of Cosines <u>Additional Terms:</u> N/A		Calculators	
	<u>Additional Preparation</u> This is an Honors lesson. Only students enrolled in Geometry Honors should be required to complete this lesson.			

 Teacher Tips	Common Misconceptions - Students may not apply the order of operations correctly when evaluating the $2ab \cos(C)$, and accidentally take the cosine of the wrong value. - Students may use the Law of Sines instead of the Law of Cosines.	Things to Consider Students often make errors with their calculations by incorrectly using their calculator. Consider the addition of a brief tutorial or think-aloud, showing students how to find the value correctly in their calculator.
	Literacy and Language Routines Think-Pair-Share 7.H2.2 Collaboration provides students with an opportunity on deriving the Law of Cosines. Students will work together using this instructional routine. Students will have the opportunity to work independently before sharing with a partner in a collaborative discussion. Students will then prepare to share their discussion with the class.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure During this lesson, students will discover the Law of Cosines and how to use it to solve mathematical and real-world problems. Through the lesson activities, students will need to focus on relevant details within a problem, and create plans and procedures to logically order steps to solve problems.
	 Differentiate Instruction Consider your groupings for the 7.H2.2 Collaboration carefully. Use the performance data from the prior lesson to determine groupings, as this Collaboration activity will unfold in a similar structure as the one in the previous lesson.	
Lesson Notes:		

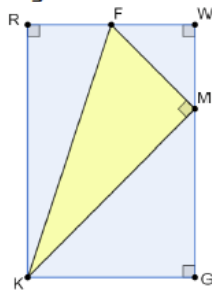
7.H2.1 Warm-Up: Reasoning With Triangles

Component Overview: Students will review a diagram of a triangle inscribed in a rectangle and answer two questions, including evaluating arguments.



7.H2.1 Warm-Up: Reasoning with Triangles

1. A diagram of a triangle inscribed in a rectangle is shown.



This activity provides students with an opportunity to engage in MA.K12.MTR.4.1: Discussion and Reflection. Students will get to analyze the mathematical thinking of others, justify results by explaining methods and processes, and construct possible arguments based on evidence.

- a. Liu Yang told her partner that all of the right triangles in the diagram are similar. Determine if the diagram contains enough information to verify Liu Yang's statement. Explain your reasoning.
Answers will vary. The diagram does not contain enough information to determine if the triangles are similar. In order to determine similarity, the image would have to show an additional angle for each image or proportional side lengths.
- b. In the diagram, $m\angle RFK = 72^\circ$ and $m\angle MKF = 63^\circ$. Use these measures to evaluate your reasoning. If necessary, make adjustments.
With this given information, $\triangle FWM \sim \triangle MGK$ as they both have angle measures of 90° , 81° , and 9° . The other two triangles pictured have different angle measures and are therefore not similar. Liu Yang's statement is still incorrect that all of the right triangles in the diagram are similar.

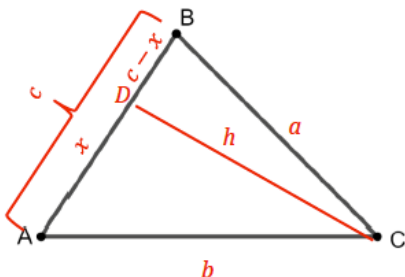
7.H2.2 Collaboration: Deriving the Law of Cosines

Component Overview: Students will work together with a partner or a small group to derive the formula for the Law of Cosines.



7.H2.2 Collaboration: Deriving the Law of Cosines

1. A triangle is shown below.



- a. Use $\triangle ABC$ to complete the following.

- Sketch an altitude from vertex C .
- Label the altitude h , and label the point on $\overline{AB}$ as D .
- Label each line segment. $AC = b$, $AD = x$, $CB = a$, $DB = c - x$, $AB = c$

The altitude creates two right triangles inside $\triangle ABC$. Notice that $\angle A$ is contained in one of the right triangles, and $\angle B$ is contained in the other.

- b. Use the labels on the two right triangles inside $\triangle ACB$ to complete the table.

$\triangle ADC$	$\triangle CDB$
$x^2 + h^2 = b^2$	$h^2 + (c - x)^2 = a^2$

- c. What do you notice about both equations?

Both equations have the variable term h^2 .



This activity provides students with an opportunity to collaborate on deriving the Law of Cosines. This can potentially cause frustration, as it requires students to think deeply and make connections, instead of having the answer up front. Students will be encouraged to build perseverance by modifying methods as needed while solving a challenging task.



What is the purpose of sketching an altitude as part of this problem?

7.H2.2 Collaboration: Deriving the Law of Cosines (continued)

- d. Rearrange the Pythagorean Theorem for $\triangle ADC$ in terms of h^2 .

$$h^2 = b^2 - x^2$$

- e. Substitute the new equation in place of h^2 in $\triangle CDB$.

$$b^2 - x^2 + (c - x)^2 = a^2$$

- f. Use cosine to write the trigonometric ratio for $\angle A$.

$$\cos A = \frac{x}{b}$$

- g. Solve the trigonometric ratio for $\angle A$ for x .

$$x = b(\cos A)$$

- h. Substitute $b \cos A$ in place of x in the equation from part e.

$$b^2 - (b(\cos A))^2 + (c - (b(\cos A)))^2 = a^2$$

- i. Expand and combine like terms.

$$a^2 = b^2 + c^2 - 2bc(\cos A)$$



Students who struggle with questions 1d through 1i may need additional support with literal equations/rearranging formulas from prior units.



Students may not recall what it means to "expand." Consider this misconception when circulating during this activity.

If students struggle with expanding from question 1h to question 1i, consider having them expand and combine like terms in question e before substituting in for question 1h.

7.H2.3 Guided Instruction: Law of Cosines

Component Overview: Students will be formally introduced to the Law of Cosines and its use in two, multi-part problems.



7.H2.3 Guided Instruction: Law of Cosines



The **Law of Cosines** can be written as one of the following:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

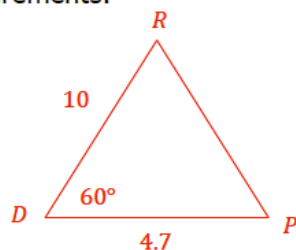
$$c^2 = a^2 + b^2 - 2ab \cos C$$



Consider creating an anchor chart that displays the Law of Cosines, along with a specific example or two of it being applied as a reference point for students to use as they complete this lesson and the remainder of the unit.

1. In $\triangle DRP$, $DR = 10$, $DP = 4.7$, and $m\angle RDP = 60^\circ$.

- a. Draw and label a triangle with the given measurements.



- b. Use the law of cosines to find the measure of $\overline{RP}$.

$$(RP)^2 = 10^2 + 4.7^2 - 2(10)(4.7)(\cos 60)$$

$$RP \approx 8.665$$

- c. Use the law of cosines to find $m\angle RPD$.

$$10^2 = 4.7^2 + 8.665^2 - 2(4.7)(8.665)(\cos P)$$

$$m\angle RPD \approx 91.990^\circ$$



This activity provides students with explicit instruction on the Law of Cosines, bridging what students discovered in the prior component with the tasks they will complete in the next component. Use this time to demonstrate thinking out loud, and prompt students to reflect and respond to that thinking.

7.H2.3 Guided Instruction: Law of Cosines (continued)

2. The measurements of the sides of $\triangle RLP$ are $RL = 4.5$, $PL = 7.2$, and $RP = 6.7$. Determine the measure of the angles of $\triangle RLP$.

Angle	Measurement
$\angle L$	65.181°
$\angle P$	37.561°
$\angle R$	77.258°

$$7.2^2 = 4.5^2 + 6.7^2 - 2(4.5)(6.7) \cdot \cos(R)$$

$$m\angle R \approx 77.258^\circ$$

$$6.7^2 = 4.5^2 + 7.2^2 - 2(4.5)(7.2) \cdot \cos(L)$$

$$m\angle L \approx 65.181^\circ$$

$$180 - (77.258 + 65.181) = 37.561$$

$$m\angle P \approx 37.561^\circ$$



Do you find the use of the Law of Cosines challenging? Why or why not?

7.H2.4 Guided Practice: Law of Cosines

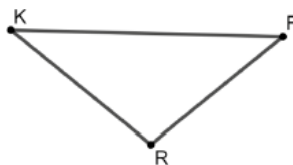
Component Overview: Students will practice two problems on the use of the Law of Cosines.



7.H2.4 Guided Practice: Law of Cosines

1. $\triangle KFR$ is shown. $KF = 15$, $FR = 9$, and $KR = 10$. Find the missing angles.

Angle	Measurement
$\angle F$	40.274°
$\angle K$	35.577°
$\angle R$	104.149°



$$15^2 = 10^2 + 9^2 - 2(10)(9)(\cos(R))$$

$$m\angle R \approx 104.149^\circ$$

$$10^2 = 15^2 + 9^2 - 2(15)(9)(\cos(F))$$

$$m\angle F \approx 40.274^\circ$$

$$180 - (104.149 + 40.274) = 35.577$$

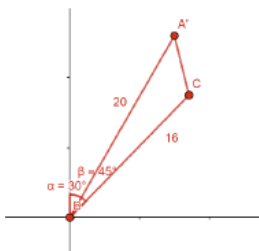
$$m\angle K \approx 35.577^\circ$$

2. Two fishing boats begin traveling in the same direction from a port in Miami. One travels 20 nautical miles per hour on a bearing of 30° and the other travels 16 nautical miles per hour on a bearing of 45° . Assuming each boat maintains its course and speed, how far apart will the fishing boats be after three hours?

$$c^2 = 20^2 + 16^2 - 2(20)(16)(\cos(15^\circ))$$

$$c \approx 6.149$$

$$3c \approx 18.447 \text{ miles apart}$$



Allow informal observation throughout the previous two components to dictate how “guided” this Guided Practice should be for you and your classroom. Consider small group instruction for some while others attempt independently, or consider teacher facilitation of practice with students actively working as a whole group.



Some students may struggle with question 2 because no visual is provided and there is a lot of information in the question. Encourage students to create a triangle and label the information provided to use as a visual for visual learners. Consider using the MLR 6: “Three Reads” routine for auditory learners to make sense of the question.

7.H2.5 Your Turn!

Component Overview: Students work independently to practice using the Law of Cosines to solve two mathematical problems.



7.H2.5 Your Turn!

1. In $\triangle BNW$, $BN = 9$, $NW = 12$, and $BW = 13$. Find the missing angles.

Angle	Measurement
$\angle B$	63.064°
$\angle N$	74.974°
$\angle W$	41.962°

$$13^2 = 9^2 + 12^2 - 2(9)(12)(\cos(N))$$

$$m\angle N \approx 74.974^\circ$$

$$12^2 = 13^2 + 9^2 - 2(13)(9)(\cos(B))$$

$$m\angle B \approx 63.064^\circ$$

$$180 - (75 + 63.1) = 41.962$$

$$m\angle W \approx 41.962^\circ$$

2. In $\triangle SLC$, $SC = 34$, $SL = 22$, and $m\angle LSC = 42^\circ$. Find all missing sides and angles.

Part	Measurement
$\overline{LC}$	22.984
$\angle C$	39.828°
$\angle L$	98.172°

$$(LC)^2 = 22^2 + 34^2 - 2(22)(34)(\cos(42))$$

$$(LC)^2 \approx 528.255$$

$$LC \approx 22.984$$

$$22^2 = 34^2 + 22.984^2 - 2(34)(22.984)(\cos(C))$$

$$m\angle C \approx 39.828^\circ$$

$$180 - (42 + 39.828) = 98.172$$

$$m\angle L \approx 98.172^\circ$$



Encourage students to create a visual by labeling the information given. This will allow students to see the connection between each given side length, and each missing angle.



- What are some similarities that you have noticed between the Law of Sines and the Law of Cosines?
- What are some of the differences?

7.H2.6 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.H2.6 Wrap-Up: Cool Down

1. Write a summary of what you learned today.

Answers will vary.